This ICCV Workshop paper is the Open Access version, provided by the Computer Vision Foundation.
Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

Inland Excess Water (IEW) Monitoring Using Sentinel-1/2: A SplitClass Segmentation and Temporal Gap-Filling Approach

Yahya Ibrahim[1], Márta Belényesi[2], Chang Liu[1], Mátyás Richter-Cserey[2]
Máté Simon[2], Tamás Szirányi[1], Csaba Benedek[1]

[1] HUN-REN Institute for Computer Science and Control (SZTAKI) [2] Lechner Knowledge Center

Emails: lastname.firstname@{sztaki.hun-ren.hu, lechnerkozpont.hu}

Abstract

Accurate mapping of Inland Excess Water (IEW), encompassing waterlogging and water retention areas, is essential for advancing sustainable land management and large-scale water conservation efforts. Existing remote sensing methods often focus on Binary water-nonwater segmentations, struggle in wet-dry transitional zones, and face challenges from cloud-related data gaps. To address these limitations, we propose a novel approach that combines what we refer to as SplitClass segmentation with a temporal Gap-Filling framework. SplitClass segmentation is an adaptive approach that assigns either a single class or a pair of top-2 class labels to each pixel, depending on prediction confidence. Meanwhile, the framework fuses temporal optical (Sentinel-2) and radar (Sentinel-1) data, ensuring improved temporal consistency and mitigating cloud occlusions. To support this research, we present IEW-Seg, a new dataset comprising fine-grained waterlogging classes, multi-source satellite inputs, and field-validated ground truth. Quantitative evaluation demonstrates that our method significantly enhances IEW detection accuracy. The official website and more detailed information are available https://waterdetection.github.io/water_detection/

1. Introduction

Climate change-induced intensification of extreme weather events presents two primary hydrological challenges: increased waterlogging, which affects soil health, crop productivity, and infrastructure stability [31, 47]; and a decline in landscape water storage capacity, exacerbating water scarcity and impacting both agricultural and natural wetland ecosystems. To address these issues, precise spatial delineation of waterlogged areas offers dual benefits: mitigating the impacts of waterlogging while promoting sustainable land management [23, 49]. Additionally, it enables

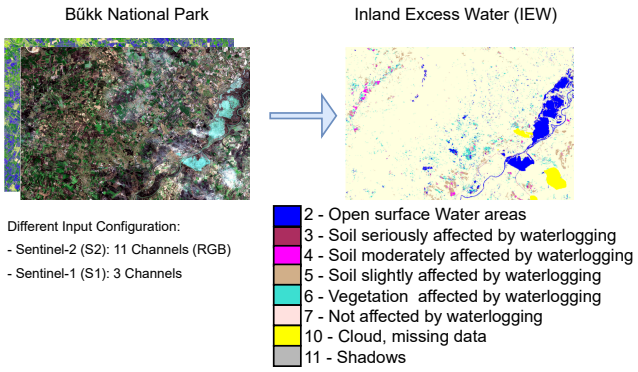


Figure 1. Overview of the IEW-Seg dataset: input channels, inland excess water (IEW) map with class IDs and colors.

the identification of natural water retention zones, which are crucial for large-scale conservation and climate adaptation. Incorporating historical water retention patterns into land management strategies enhances sustainability and aligns with EU frameworks for climate resilience and water resource governance [12, 56, 57].

Remote sensing (RS) technologies, including Artificial Intelligence (AI) powered optical and radar satellite imagery, have proven to be powerful tools for detecting water bodies in diverse landscapes [21, 62]. While existing methods effectively map open water bodies [4, 26], they struggle to accurately capture the varying degrees of water impact on soil and vegetation—critical for understanding Inland Excess Water (IEW) dynamics [53], particularly in transitional zones.

The 100-square-meter pixel size of optical satellite data makes it challenging to capture subtle hydrological changes. Gradual transitions from wet to dry conditions, passing through various degrees of severity, are difficult to map accurately. In addition, persistent cloud cover presents a significant challenge, substantially limiting the availability of high-resolution optical imagery. Although radar

can partially overcome this bottleneck, C-band sensors like Sentinel-1 remain ineffective in detecting the subtle variations in moisture content across soil and vegetation, limiting its ability to provide a comprehensive assessment of IEW.

Traditional single-class segmentation methods are limited by their rigid classifications, which often fail to accurately represent areas with mixed or ambiguous characteristics. To address this limitation, we propose a dynamic soft segmentation strategy, which we term *SplitClass* Segmentation, offering a practical alternative to full soft-label segmentation, designed to better represent ambiguous or mixed-class regions in large-scale geospatial data. Instead of assigning full probability distributions, our method assigns either a single class in confident areas or a combination of the top-2 most likely classes in uncertain regions. These two-class assignments are treated as newly defined composite classes — each representing a unique pairwise combination of base classes — and are visually represented in the segmentation map using blended colors, typically created by mixing RGB values of the two original classes in equal proportions. This approach provides a more intuitive and user-friendly interpretation, especially in high-resolution maps. We use a knowledge distillation framework, where a student model learns from the soft outputs of a teacher to estimate class probabilities. However, instead of storing and visualizing full soft maps, SplitClass Segmentation offers a scalable and interpretable compromise by dynamically switching between single and dual-class outputs based on prediction confidence.

Furthermore, we propose a temporal gap-filling approach that improves temporal consistency and integrates cross-modal data. In this framework, gaps in Sentinel-2 (S2) data caused by cloud cover are filled using Sentinel-1 (S1) radar-based water detection. Additionally, observations from cloud-free S2 images before and after the missing data period are leveraged to maintain consistency, ensuring a more reliable and temporally stable representation of waterlogging patterns.

To validate our approach, we introduce IEW-Seg a publicly available dataset specifically designed for the detection of IEW. This dataset integrates multi-source input data, including S2 optical imagery and S1 radar data. IEW-Seg provides a detailed segmentation map that categorizes distinct classes: water bodies, vegetation affected by water, and soil affected by varying degrees of waterlogging severity. The dataset is derived from two national park lands, with training and validation data temporally separated across the 2020–2024 period, ensuring its ability to generalize to unseen waterlogging conditions and capture their dynamic evolution over time. The dataset provides field data collected in 2024, where experts classified parts of the studied lands on specific testing dates. These ground-truth observations were synchronized with remote sensing imagery to assess the accuracy of the proposed strategy, thereby demonstrating its practical applicability.

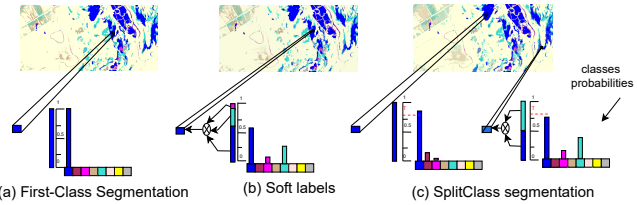


Figure 2. Overview of Output Segmentation Representations: (a) single-class prediction, (b) soft label probabilities, and (c) our SplitClass method blending two classes under low confidence (threshold indicated by a dashed red line).

Fig. 1 illustrates our dataset samples, with labeled segmentation classes and assigned IDs. While Fig. 2 illustrates three different representations of the output segmentation: (a) the single-class segmentation, where the teacher model is trained to predict the most probable class; (b) the soft labels, where we show the segmentation image by blending class colors based on their respective probabilities. In most cases, the dominant first class typically appears prominently due to its high probabilities; and (c) *SplitClass* segmentation, where a single class is assigned when the model's confidence is high. When the prediction is uncertain, a blended representation of two plausible classes, each contributing equally, is used for visualization.

The key contributions of this work are as follows:

- SplitClass segmentation for large-scale imagery: We propose a segmentation strategy using top-2 class predictions in low-confidence areas to boost reliability on large-scale satellite data.
- Temporal gap-filling: We integrate the segmentation strategy into a temporal gap-filling framework to address missing optical data.
- A Novel Dataset for Waterlogging Detection: We introduce IEW-Seg, a specialized dataset that characterizes waterlogged areas with fine-grained classes, enabling more precise segmentation.

2. Related Work

2.1. Water Segmentation Datasets and Methods

The detection of surface water using remote sensing imagery has been extensively researched, leading to the development of numerous datasets and methodologies. These approaches primarily rely on data from S1 Synthetic Aperture Radar (SAR), S2 optical imagery, or a fusion of both modalities.

SAR imagery is particularly valuable due to its ability to penetrate cloud cover and adverse weather conditions [42], making it widely used in the literature for water body detection. Notable Sen1Floods11 [7] detects flooded open areas

at 11 flood events, MM-Flood [34] supports multi-modal flood mapping. The UrbanSARFloods dataset [70] captures 18 global flood events, while UW-Bench [50] addresses diverse urban flood scenarios. Additionally, RAPID-NRT [63] enhances near-real-time flood detection by integrating topography and historical water occurrence data. Optical datasets based on S2 imagery have also been widely explored. SpaceNet 8 [20] offers high-resolution optical data for identifying flooded buildings and roads. FloodNet [41] provides airborne optical imagery tailored for urban flood scenarios, while studies such as [13] focus on leak detection in urban areas. The GloLakes dataset [19] enables historical and near-real-time monitoring of global lake water storage. Recent datasets using both S1 and S2 imagery include OptiSAR-POM [29], which applies an object-based method to detect small ponds, and S1S2-Water [60], offering reference data for surface water mapping.

Traditional water detection methods often rely on thresholding applied to spectral indices such as the Normalized Difference Water Index (NDWI) [15], Modified NDWI (MNDWI) [1], Water in Wetlands Index (WIW) [25], Augmented NDWI (ANDWI) [40], and Normalized Difference Photovoltaic Index (NDPI) [43]. However, these approaches are sensitive to environmental conditions, which can result in inconsistent outputs. To enhance robustness, probabilistic models like Fusion Markov Random Fields (f-MRF) [48, 52] have been explored, incorporating spatial and temporal dependencies. Yet, these methods require manually defined energy functions and are computationally intensive. With the rise of machine learning, classifiers such as Support Vector Machines (SVM) [24] and Random Forests (RF) [16] have been applied. Although these models generalize better, they still rely on handcrafted features and often struggle in complex scenes. Recent advances in deep learning have significantly improved surface water detection and segmentation. A shallow Convolutional Neural Network (CNN) architecture followed by fully connected layers is utilized for binary lake classification [13], while a VGG19-based fully convolutional network with transfer learning is applied for similar tasks [61]. Several state-of-the-art CNNs have also been evaluated for water segmentation tasks [70]. Other studies tackle data scarcity by using proxy-based label collection methods [35].

Recent methods have demonstrated strong performance, yet they often struggle to accurately segment transitional or mixed-class pixels, an inherent challenge in remote sensing caused by coarse spatial resolution. To address this, soft segmentation approaches that produce class probability distributions have gained popularity. Such methods have shown promise in both medical imaging [30] and remote sensing applications [55, 59]. These methods are considered fuzzy because they allow each pixel to belong partially to multiple classes, making them useful for representing continuous landscapes where discrete class boundaries are unclear [14]. Nevertheless, most existing works in water detection remain focused on binary water classification (i.e., water vs. non-water) and are ill-suited for detecting IEW or assessing the severity of waterlogging. To bridge this gap, we introduce IEW-Seg, a novel dataset designed for detailed classification of waterlogging severity over time, along with a segmentation output representation that aligns with its fine-grained labeling scheme. This representation is well-suited for large-scale remote sensing imagery and provides an effective means of capturing uncertainty and gradual transitions in waterlogged areas.

2.2. Knowledge Distillation in Remote Sensing

Knowledge Distillation (KD) was originally proposed in [18] as a technique for transferring knowledge from a large, complex teacher model to a smaller, more efficient student model. KD methods are commonly classified into three categories. Probability-based KD [18] transfers softened output probabilities from the teacher to the student. Feature-based KD [44] aligns intermediate feature representations. Relation-based KD preserves structural relationships between data samples, such as pairwise similarities [39]. KD has gained significant traction in remote sensing as an effective approach to compress models while preserving high performance [17]. It has been successfully applied to various tasks, including scene classification [22] and change detection [58], by combining prototypical contrastive learning with channel-spatial normalization. For semantic segmentation, KD techniques employ graph attention guidance [71] and cross-model distillation between CNN and transformer-based architectures to leverage their complementary strengths [11]. Additionally, self-supervised KD frameworks have been developed to address few-shot learning scenarios [68].

In the domain of water segmentation, KD has been effectively employed to enhance both model accuracy and computational efficiency. KD is leveraged to improve water segmentation performance through a multimodal fusion framework [66]. Similarly, KD is applied within a CNN-based approach for sea ice segmentation [64]. While most KD methods focus on improving student model efficiency and real-time performance by imitating the teacher's soft labels, we explore an adaptation of the teacher-student framework that leverages these soft labels to enable a SplitClass representation. By capturing the top-2 predicted classes, this approach preserves richer decision information and enhances robustness in complex remote sensing scenarios such as mixed land-water areas.

3. IEW Monitoring Pipeline

3.1. Dataset Construction and Annotation

Effective water management is crucial in Hungary, where over 70% of the country has experienced droughts in the past decade, while up to 15% of arable land has periodic flooding, and the areas affected by both phenomena overlap significantly [6]. This dual challenge is present in the areas we specifically studied: the Heves Steppes (located in the Bükk National Park, referred to as BNPI) and the sodic lakes area in the Upper-Kiskunság (located in the Kiskunság National Park, referred to as KNPI).

3.1.1. Dataset Composition and Experimental Design

The dataset comprises multi-temporal satellite imagery collected from both S2 and S1 missions. From S2, we utilize 11 out of the 13 available spectral bands, excluding bands B9 and B10 due to their limited relevance for our analysis. S1 contributes Sigma nought (Sigma-0) backscatter coefficients (VV, VH, and their ratio), calculated using the Hungarian Earth Observation Information System [33]. These radar-based measurements are particularly effective for distinguishing water surfaces from land [5] and have been widely used in hydrological studies [3, 32]. Due to differences in revisit cycles, S1 and S2 images were not acquired on the exact same dates; however, the temporal difference typically did not exceed ± 2 days. For both studied regions, the dataset includes multi-year, multi-season observations covering consistent geographical areas, enabling the model to capture diverse water conditions and seasonal variation. The dataset is split into training and testing sets for model evaluation. Training sets for BNPI and KNPI each include 7 samples. Testing sets consist of 4 samples for BNPI and 5 samples for KNPI, with ground truth data for quantitative performance assessment. For the KNPI dataset, data before 2023 was used for training, with subsequent samples reserved for testing to ensure temporal separation. In contrast, for the BNPI dataset, the test set includes earlier dates to assess the model's generalization capability across different time periods (Acquisition dates and data details are in the Supplementary Material).

3.1.2. Inland Excess Water (IEW) Map Generation

The generation of IEW ground truth maps (see Fig. 1) is a critical step for developing supervised models and evaluating their performance. These maps offer a detailed representation of waterlogging dynamics, enabling precise segmentation. This process involved a multi-step, expert-supervised process using an improved version of a previously developed methodology [36]:

- *Initial IEW Map Creation:* S2 Level-2A (L2A) images, which provide atmospherically corrected surface reflectance (Bottom-of-Atmosphere, BOA), were analyzed. Visual interpretation and expert-defined, per-image thresholding were applied to Short Wave Infrared band (SWIR1) along with three spectral indices: the Normalized Difference Vegetation Index (NDVI) [46], and the Normalized Difference Moisture Index (NDMI) [8], and a custom index defined as $(\text{SWIR1} - \text{RED})/(\text{SWIR1} + \text{RED})$ applied operationally within the Lechner Knowledge Center, SWIR1 and RED are sensitive to soil moisture and useful for mapping bare soil water status [65]. This step identified six thematic classes: open water, soils with varying water impact (seriously, moderately, and slightly), vegetation affected by water, and unaffected areas.
- *Masking Built-up and Forested Areas:* Built-up and forested areas were masked using the Hungarian high-resolution land cover layer, further refining the initial IEW maps.
- *Cloud and Shadow Mapping:* Clouds and shadows were mapped using S2 scene classification maps, with manual corrections applied to improve accuracy.
- *Final IEW Map Integration:* The cloud and shadow layers were integrated into the refined maps, resulting in the final IEW maps.

3.1.3. Field Data Collection

To validate the remote sensing-based IEW detection, we extended the testing dataset with field observations collected throughout 2024. These observations span different seasons and locations across the study areas, ensuring comprehensive coverage of diverse hydrological conditions. Field data collection was conducted using the QField application [38], where experts documented various parameters for each sample point, including the date of collection, a unique ID, GPS coordinates (x-coordinate, y-coordinate), and the assigned class for IEW status. Additionally, water depth, wet area, vegetation cover, and vegetation height were measured to provide comprehensive on-site data. To further enhance the dataset, a photo was taken at each sample point from a single direction, with the camera's orientation recorded. This directional information is critical for aligning field observations with satellite imagery, as the GPS coordinates and camera orientation are synchronized with the satellite images to ensure accurate spatial matching, enabling reliable comparison between ground-based and satellite data.

For BNPI, data was collected over 9 days throughout the year, resulting in 422 samples. For KNPI, data was collected over 12 days, totaling 694 samples. Fig. 3 displays the spatial distribution of samples in KNPI, with arrows in different colors representing samples collected on the same day. This visualization highlights the temporal and spatial diversity of the field data, which is essential for robust model evaluation.

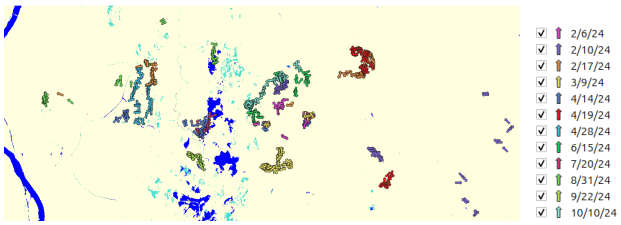


Figure 3. Field data For KNPI land, with each arrow color representing samples collected on the same day in 2024, the background color highlights the land for clarity.

3.2. Proposed Method

The objective of the proposed method is to develop an automated, adaptive method for detecting and differentiating waterlogged soil and vegetation to support decision-making. We propose a dual-model strategy to address the challenges posed by varying cloud conditions in satellite imagery. For cloud-free scenarios, we utilize S2 input data based on a teacher-student framework, where the teacher model is trained directly on the hard ground truth labels, with the aim of producing soft, informative predictions that capture richer inter-class relationships and underlying uncertainties. Since manually generating such soft labels is impractical, we use the teacher's probabilistic outputs to supervise the training of the student model (referred to as `S2_Model`). This guidance enables the student to learn more nuanced feature representations and enhances its generalization capabilities. We employ a self-distillation approach in which both the teacher and student models share the same architecture.

Under cloudy conditions, we employ a radar-based model `Radar_Model` trained on S1 channels to detect waterlogged areas, leveraging radar's ability to penetrate cloud cover. During training, regions labeled as clouds or shadows in the ground truth required special handling. These labels were derived from optical S2 imagery, which cannot observe surface conditions beneath cloud cover or shadows—leaving the water status in such areas unknown. As these regions lack reliable supervision, they were excluded from the loss computation by preventing gradient backpropagation, effectively omitting them from the learning process to avoid introducing uncertainty into the model. Furthermore, radar backscatter depends on surface properties such as roughness and moisture, causing waterlogged soil and vegetation to have similar values. To minimize the chance of misclassification, a binary classification approach was adopted, where all water-affected classes were consolidated into a single 'water' class, simplifying the model to categorize regions as either water or non-water.

Both models share an identical architecture, differing only in the number of input channels, which are adjusted to match the specific input data. Each scenario introduces distinct challenges, which are addressed through a combination of KD, SplitClass segmentation, and a temporal gap-filling framework.

3.2.1. SplitClass Segmentation

In cloud-free regions, ambiguous situations often occur—particularly in areas with mixed land cover or where transitions happen at the edges of water bodies. To address such cases during inference, we propose a SplitClass segmentation approach. In our method, if the model is confident in its prediction, we present only the most probable (Top-1) class for each pixel. However, when the probability of the top predicted class falls below a predefined threshold, we retain and present the two most probable classes (Top-2) instead of forcing a single decision. This method is designed to provide a clearer representation of uncertainty, especially in high-resolution remote sensing imagery, where visualizing the top two classes can help end-users better interpret transitional or ambiguous areas. Details of the implementation are provided in Algorithm 1. This dual-class assignment introduces a new composite class that represents the coexistence or ambiguity between two base categories. Importantly, the number of these additional composite classes remains limited and computationally manageable: for n base classes, the number of possible Top-2 combinations is given by the binomial coefficient $C(n, 2)$. In our case, with 6 base classes (water-related classes), this produces only 15 additional composite classes.

Algorithm 1 SplitClass Segmentation

for each pixel x_i **do**
  $C_{\text{top-1}} \leftarrow \arg\max_k P(x_i, C_k)$
  if $P(x_i, C_{\text{top-1}}) \geq$ threshold **then**
    $x_i \leftarrow C_{\text{top-1}}$
  else
    $C_{\text{top-2}} \leftarrow \arg\max_{k \neq C_{\text{top-1}}} P(x_i, C_k)$
    $x_i \leftarrow (C_{\text{top-1}}, C_{\text{top-2}})$ ▷ Assign both classes
  end if
end for

3.2.2. Temporal Gap-Filling Approach

During the testing phase, the `S2_Model` is initially applied to predict water presence and severity in cloud-free observations while simultaneously identifying cloud-covered regions. In the presence of clouds, the `Radar_Model` is used to detect waterlogged areas, as optical imagery is unavailable. However, due to its binary nature, this model may lack the granularity needed for reliable segmentation. To address this limitation, we incorporate information from the previous and next cloud-free observations of the `S2_Model` to support interpretation in the cloudy region. Specifically, if a given area is classified as waterlogged in both the preceding

and subsequent cloud-free observations, it is inferred to be waterlogged during the cloudy period as well. Furthermore, to capture gradual changes in water severity, we apply an adapted SplitClass segmentation strategy: the top predicted class from the previous cloud-free observation and the top predicted class from the next one are both assigned to the corresponding pixel in the region under analysis. Details are provided in Algorithm 2.

Algorithm 2 Temporal Gap-Filling approach

Require: $s2_prev$: Previous S2 result, $s2_next$: Next S2 result, $s2_day$: S2 result of the studied day, $radar_mask$: Radar water detection of the studied day. "Water" refers to Open Surface Water (class 2).

$mask \leftarrow (s2_day = \text{missing})$
$s2_day[mask \wedge (radar_mask = \text{water})] \leftarrow$ water
Update $s2_day$ based on time series:
if $s2_prev = s2_next$ **then**
    $s2_day \leftarrow s2_prev$
else if Either $s2_prev$ or $s2_next$ is missing **then**
    $s2_day \leftarrow$ available scan
else
    $s2_day \leftarrow (s2_prev, s2_next)$ ▷ Assign both classes
end if
return $s2_day$

4. Experiments and Results

4.1. Implementation details:

The imagery resolution varies across the studied areas, with (5400×3600) pixels for BNPI, and (4800×2400) pixels for KNPI. In the training setup, we used an image size of 64×64 pixels, with a batch size of 32, trained over 100 epochs, and a learning rate of 10^{-4} to optimize the model's performance.The experiments were carried out on a personal computer featuring an AMD Ryzen 9 5900X 12-Core Processor, 32 GB of RAM, and an NVIDIA GeForce RTX 3060 Ti GPU.

4.2. Evaluation of DNN-Based Models

The objective of this experiment is to identify the most effective teacher model for training student models. To this end, we performed our experiments within the framework of top-1 class segmentation. We conducted a performance analysis of state-of-the-art segmentation models U-Net [45], U-Net++ [72], FPN [27], PSPNet [69], DeepLabv3 [10], MANet [2], Attention UNet [37], ResidualAttentionUNet [37], TransUNet [9] and RUPNet [54]. Training was performed on S2 imagery using a combination of Weighted Focal Loss (WFL) [28] and Generalized Dice Loss (GDL) [51], where class weights were determined based on the frequency of each class in the training data to address class imbalance.

The models' performance was evaluated using key classification metrics as in [13]: Precision (Pr), Recall (Rc), and F1 Score (F1), for the five key classes: open surface water, soil (seriously, moderately, and slightly affected by waterlogging), and vegetation affected by water. The results presented in Tab. 1 show the performance of each model. RUPNet [54] achieved the highest F1 score, demonstrating a strong balance between precision and recall across all models. This highlights its capability to accurately segment small to medium-sized regions, making it particularly well-suited for remote sensing-based detection of waterlogged areas. As a result, we decided to proceed with this model in the next experiments.

Model	**Pr** (%)↑	**Rc** (%)↑	**F1** (%) ↑
U-Net [45]	78.58	80.20	79.38
U-Net++ [72]	77.61	80.87	79.10
FPN [27]	73.31	72.55	72.93
PSPNet [69]	69.79	69.13	69.46
DeepLabv3+ [10]	72.29	73.72	73.00
MANet [2]	77.80	80.91	79.32
AttentionUNet [37]	**78.40.**	77.06	77.77
ResidualAttentionUNet [37]	77.67	78.27	77.97
TransUNet [9]	75.89	77.86	76.86
RUPNet [54]	76.35	**86.70**	**81.19**

Table 1. Model Performance Comparison on the IEW-Net Dataset: Evaluation of DNNs for Water-Related Classes (Classes 2–6).

4.3. Evaluation of SplitClass Segmentation

RUPNet [54] was selected as the teacher model for knowledge distillation. A student model with the same architecture was trained using a combination of GDL loss, WFL loss, and Kullback–Leibler (KL) divergence loss to align the student's output distribution with that of the pre-trained teacher. After training, we applied the Splitclass segmentation as a post-processing step on the student model's output.

Table 2 presents a quantitative evaluation of the teacher, student as top class segmentation, as well as with the proposed SplitClass approach for each class and overall, weighted by the number of pixels per class, where a prediction was considered a true positive if either of the SplitClass predicted classes matched the ground truth. The results are compared against standard (hard) segmentation on the testing dataset using precision, recall, and F1-score.

The student model slightly outperforms the teacher. This observation supports the idea that students trained with distillation objectives can generalize better than their teachers, as previously demonstrated in [67], where self-distillation led to improved performance despite no architectural changes. Furthermore, applying SplitClass segmen-

tation to the student's predictions further enhances the results. We present the results using a threshold of $\theta = 0.98$, as this setting provides a balanced trade-off between improving accuracy for underrepresented classes and the proportion of pixels assigned dual-class (SplitClass) labeling. Additional threshold analyses are provided in the supplementary material. Under this threshold, SplitClass segmentation affects a relatively small percentage of pixels overall (6.33%), yet it yields improvements in F1-score for underrepresented classes (3–6), which are often difficult to distinguish due to overlapping spectral characteristics. These underrepresented classes are often difficult to distinguish due to overlapping spectral characteristics. A small fraction of pixels (0.13%) involve the merging of opposite classes (water bodies and unaffected areas), primarily in highly ambiguous boundary regions. This minor mismatching reflects inherent challenges in remote sensing and does not diminish the overall performance of the proposed method. A more notable observation is the significant overlap between the "vegetation affected by water" and "non-affected areas" classes (2.33%). This can be attributed to the challenge of accurately segmenting wet vegetation that is not fully submerged, as it often shares visual characteristics with non-waterlogged vegetation regions.

ID	Teacher Top1			Student Top1			SplitClass $\tau = 0.98$		
	Pr ↑	Rc ↑	F1 ↑	Pr ↑	Rc ↑	F1 ↑	Pr ↑	Rc ↑	F1 ↑
2	0.98	0.99	0.98	0.98	0.99	0.99	0.99	1.00	1.00
3	0.70	0.60	0.65	0.72	0.62	0.67	0.91	0.90	0.90
4	0.59	0.57	0.58	0.61	0.55	0.58	0.90	0.84	0.87
5	0.53	0.76	0.63	0.57	0.67	0.62	0.91	0.90	0.90
6	0.56	0.84	0.67	0.76	0.77	0.77	0.93	0.94	0.94
7	0.99	0.97	0.98	0.99	0.98	0.99	1.00	1.00	1.00
All	0.98	0.96	0.97	0.98	0.97	0.98	0.99	0.99	0.99

Table 2. Performance Comparison of hard Segmentation for Teacher and Student model, and SplitClass Segmentation for Waterlogging Detection, by class. blue for the best, orange for the second best.

4.4. Temporal Gap-Filling Analysis

To enhance IEW detection, we integrate S2 and S1. The `S2_Model` processes cloud-free S2 data. In cloudy areas, we employ the `Radar_Model`, incorporating temporal information from nearby cloud-free S2 observations to improve reliability. This multimodal and temporal approach is illustrated in Fig. 4. The figure shows: the first row shows S2 (RGB channels) for three consecutive scans, the second row shows `S2_Model` predictions with the associated colors defined in Fig. 1, highlighting cloud cover limitations, the third row shows S1 input data for the studied day, the `Radar_Model` result, and the final combined output, demonstrating the integrating of `Radar_Model` result and temporal coherence from `S2_Model` (High-res in Suppl. Fig. S6). Open surface water areas, undetected by radar alone, were identified using the proposed methodology. This was possible due to the fact that they are water bodies in both previous and subsequent non-cloudy scans, even with varying intervals between scans. We observed that the spatial extent of waterlogged areas changed only slightly between the available scans before and after the study day, while the severity of waterlogging varied. This highlights the presence of mixed classes on that day. The method aims to generate results for cloud-covered regions; however, cloud cover prevents the reliable generation of ground truth data for quantitative analysis. Therefore, we rely on field data to evaluate the accuracy of the approach.

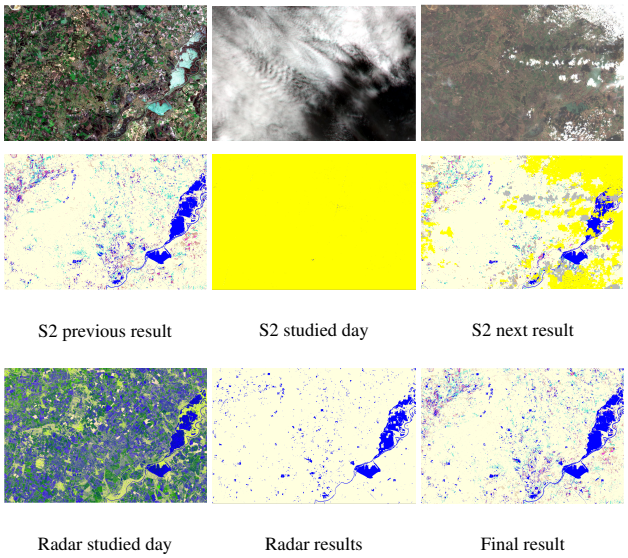


Figure 4. Multimodal and Temporal Analysis for Waterlogging Detection: Sentinel-2 RGB Images (Row 1), Model Predictions on S2 Data (Row 2), Radar Image, Radar Result, and Combined Temporal 2 Class Predictions and Radar Result (Row 3).

4.5. Real-World Validation

The performance of the proposed methodology, enhanced by Temporal Gap-Filling processing, is evaluated by comparing its results with field data collected by experts. Table 3 presents the accuracy percentages of different prediction approaches, including single-day model predictions — where the `S2_Model` is applied on cloud-free days and the `Radar_Model` is used in cloudy regions— and the results improved by Temporal Gap-Filling technology, for the KNPI and BNPI datasets. Here, accuracy refers to the proportion of correctly predicted field data points relative to the total number of collected samples. Results are presented for all samples excluding areas identified as "water under vegetation" and those smaller than 100 m^2. This filtering reduced the BNPI sample size to 321 and the KNPI sample size to 439.

Single-day predictions exhibited low accuracy, primarily due to cloudy conditions during field data collection. Under such conditions, predictions relied only on the binary water classification `Radar_Model` results. Accuracy gains are observed with Temporal Gap-Filling predictions, as incorporating past and future water status helps capture the temporal dynamics of water-affected areas. This approach enhances prediction quality and could be further improved by integrating additional factors such as weather and rainfall data in future work.

Method	BNPI (%)	KNPI (%)
Single-Day	58.25	59.45
With Temporal Gap-Filling	**72.27**	**69.24**

Table 3. Accuracy comparison of various prediction methods against field data and for KNPI and BNPI.

Fig. 5, first row, presents a side-by-side visualization: on the left is the RGB image of the studied area, and on the right is the corresponding model output (High-res in Suppl. Fig. S7). In both images, arrows indicate sample locations, each annotated with a unique ID and the expert-assigned class from field visits. The background in the right image is color-coded according to the model's predicted classes, following the color scheme and class IDs defined in Fig. 1. The second row presents field photographs illustrating actual land conditions, each linked to its corresponding ID and class. The results demonstrate that the method effectively identifies open water bodies and captures transitions from waterlogged to drier soil regions. However, some confusion is observed in detecting class 6 (vegetation affected by water), particularly when water is obscured beneath dense vegetation, which remains a challenging case.

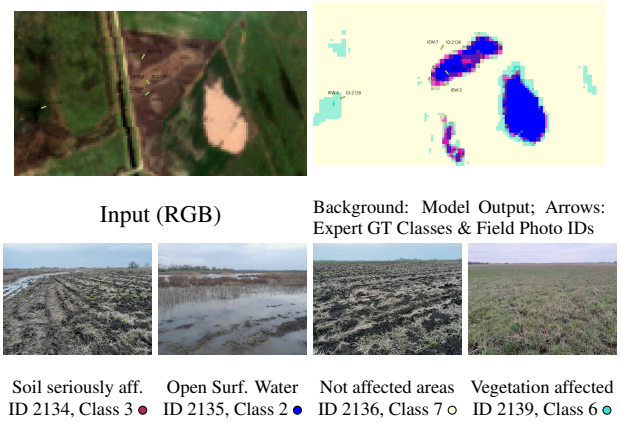


Figure 5. Field photograph showing land conditions at BNPI on March 10, 2024, with our model's predictions displayed in the background.

4.6. Cross-site Evaluation and Generalization

To demonstrate the generalizability of our method beyond the studied areas, we applied our pretrained model to a distinct and ecologically different site: the Doñana National Park in Spain. This region has been analyzed using the WIW methodology [25], providing a valuable reference for comparative evaluation. Figure 6 illustrates the input data presented in RGB color composite for the date of June 1st, 2019. The figure showcases the segmentation results produced by the WIW method, our model's Top-1 class segmentation, and the proposed SplitClass segmentation approach (High-res in Suppl. Fig. S8). Our results show that the model generalizes well to new regions. In Doñana, rice fields were accurately detected as vegetation affected by water, while parts were correctly classified as open water. Notably, several areas identified by our method as vegetation affected by water were considered non-water regions in the WIW results. The road between fields, barely visible in WIW, was clearly segmented, with some sections plausibly labeled as waterlogged soil.

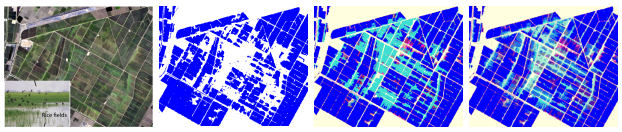


Figure 6. Waterlogging Detection in Unseen Rice Field near Doñana, Spain (June 1, 2019): Comparison of WIW Method [25], Ours Top-1 Prediction, and Ours SplitClass Segmentation. Open Surf. Water ●, Vegetation affected ●, SplitClass Water& Vegetation ●

5. Conclusion

This study has addressed critical limitations in inland excess water (IEW) monitoring by introducing a novel multi-source remote sensing data fusion approach. This method effectively mitigates challenges such as low identification accuracy in transitional zones and cloud-induced data gaps, achieving significant improvements in waterlogging detection through the integration of SplitClass segmentation and a Temporal Gap-filling framework. The efficacy of the method was further supported by the creation and validation of the IEW-Seg dataset. Quantitative evaluations, demonstrating an increase in accuracy improvement, confirm the efficacy of our approach. This study offers a robust and scalable solution for IEW monitoring, supporting climate-adaptive land management and sustainable water resource planning.

Acknowledgment

This work was supported in part by the European Space Agency Co. 4000142610, by the Hungarian Artificial Intelligence National Laboratory (MILAB, RRF-2.3.1-21-2022-00004), and also by the OTKA #143274 project of the Hungarian NRDI Office.

References

[1] Hanqiu Xu and. Modification of normalised difference water index (NDWI) to enhance open water features in remotely sensed imagery. *International Journal of Remote Sensing*, 27(14):3025–3033, 2006. 3

[2] Jingjing Bai and Yoshinobu Kawahara. MANet: Mixed Attention Network for Visual Explanation. *New Generation Computing*, 42(4):825–843, 2024. 6

[3] Bernhard Bauer-Marschallinger, Senmao Cao, Claudio Navacchi, Vahid Freeman, Felix Reuß, Dirk Geudtner, Björn Rommen, Francisco Ceba Vega, Paul Snoeij, Evert Attema, Christoph Reimer, and Wolfgang Wagner. The normalised Sentinel-1 Global Backscatter Model, mapping Earth's land surface with C-band microwaves. *Scientific Data*, 8(1):277, 2021. 4

[4] TV Bijeesh and KN Narasimhamurthy. Surface water detection and delineation using remote sensing images: A review of methods and algorithms. *Sustainable Water Resources Management*, 6(4):68, 2020. 1

[5] Filsa Bioresita, Anne Puissant, André Stumpf, and Jean-Philippe Malet. A Method for Automatic and Rapid Mapping of Water Surfaces from Sentinel-1 Imagery. *Remote Sensing*, 10(2), 2018. 4

[6] Edina Birinyi, Boglárka O Lakatos, Márta Belényesi, Dániel Kristóf, Zsolt Hetesi, László Mrekva, and Gábor Mikus. Contribution of data-driven methods to risk reduction and climate change adaptation in hungary and beyond. *Quarterly Journal of the Hungarian Meteorological Service*, 127 (4):421–446, 2023. 4

[7] Derrick Bonafilia, Beth Tellman, Tyler Anderson, and Erica Issenberg. Sen1floods11: A georeferenced dataset to train and test deep learning flood algorithms for sentinel-1. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pages 210–211, 2020. 2

[8] Bo cai Gao. NDWI—A normalized difference water index for remote sensing of vegetation liquid water from space. *Remote Sensing of Environment*, 58(3):257–266, 1996. 4

[9] Jieneng Chen, Jieru Mei, Xianhang Li, Yongyi Lu, Qihang Yu, Qingyue Wei, Xiangde Luo, Yutong Xie, Ehsan Adeli, Yan Wang, et al. TransUNet: Rethinking the U-Net architecture design for medical image segmentation through the lens of transformers. *Medical Image Analysis*, page 103280, 2024. 6

[10] Liang-Chieh Chen, Yukun Zhu, George Papandreou, Florian Schroff, and Hartwig Adam. Encoder-Decoder with Atrous Separable Convolution for Semantic Image Segmentation. In *ECCV*, 2018. 6

[11] Zhe Dong, Guoming Gao, Tianzhu Liu, Yanfeng Gu, and Xiangrong Zhang. Distilling Segmenters From CNNs and Transformers for Remote Sensing Images' Semantic Segmentation. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–14, 2023. 3

[12] Directorate-General for Environment European Commission. *EU Biodiversity Strategy for 2030 – Bringing Nature Back into Our Lives*. Publications Office of the European Union, 2021. Accessed: 2025-05-22. 1

[13] Erick Fajardo and Daniela Moctezuma. Water leak detection through satellite imagery and deep learning. *Sustainable Water Resources Management*, 11(2):33, 2025. 3, 6

[14] Giles M. Foody. Status of land cover classification accuracy assessment. *Remote Sensing of Environment*, 80(1):185–201, 2002. 3

[15] Qiandong Guo, Ruiliang Pu, Jialin Li, and Jun Cheng. A weighted normalized difference water index for water extraction using landsat imagery. *International journal of remote sensing*, 38(19):5430–5445, 2017. 3

[16] Feifei Han, Jingshan Yu, Guihuan Zhou, Shuang Li, and Tong Sun. A comparative study on urban waterlogging susceptibility assessment based on multiple data-driven models. *Journal of environmental management*, 360:121166, 2024. 3

[17] Yassine Himeur, Nour Aburaed, Omar Elharrouss, Iraklis Varlamis, Shadi Atalla, Wathiq Mansoor, and Hussain Al-Ahmad. Applications of knowledge distillation in remote sensing: A survey. *Information Fusion*, 115:102742, 2025. 3

[18] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the Knowledge in a Neural Network, 2015. 3

[19] J. Hou, A. I. J. M. Van Dijk, L. J. Renzullo, and P. R. Larraondo. GloLakes: water storage dynamics for 27 000 lakes globally from 1984 to present derived from satellite altimetry and optical imaging. *Earth System Science Data*, 16(1): 201–218, 2024. 3

[20] Ronny Hänsch, Jacob Arndt, Dalton Lunga, Matthew Gibb, Tyler Pedelose, Arnold Boedihardjo, Desiree Petrie, and Todd M. Bacastow. SpaceNet 8 - The Detection of Flooded Roads and Buildings. In *2022 IEEE/CVF CVPRW*, pages 1471–1479, 2022. 3

[21] Bhargavi Janga, Gokul Prathin Asamani, Ziheng Sun, and Nicoleta Cristea. A review of practical ai for remote sensing in earth sciences. *Remote Sensing*, 15(16):4112, 2023. 1

[22] Jinsheng Ji, Xiaoming Xi, Xiankai Lu, Yiyou Guo, and Huan Xie. From Coarse to Fine: Knowledge Distillation for Remote Sensing Scene Classification. In *IGARSS 2023 - 2023 IEEE International Geoscience and Remote Sensing Symposium*, pages 5427–5430, 2023. 3

[23] Gurpreet Kaur, Gurbir Singh, Peter P Motavalli, Kelly A Nelson, John M Orlowski, and Bobby R Golden. Impacts and management strategies for crop production in waterlogged or flooded soils: A review. *Agronomy Journal*, 112 (3):1475–1501, 2020. 1

[24] Katelyn Kirby, Sean Ferguson, Colin D Rennie, Julien Cousineau, and Ioan Nistor. Identification of the best method for detecting surface water in Sentinel-2 multispectral satellite imagery. *Remote Sensing Applications: Society and Environment*, 36:101367, 2024. 3

[25] Gaëtan Lefebvre, Aurélie Davranche, Loïc Willm, Julie Campagna, Lauren Redmond, Clément Merle, Anis Guelmami, and Brigitte Poulin. Introducing WIW for detecting the presence of water in wetlands with landsat and sentinel satellites. *Remote sensing*, 11(19):2210, 2019. 3, 8

[26] Yansheng Li, Bo Dang, Yongjun Zhang, and Zhenhong Du. Water body classification from high-resolution optical remote sensing imagery: Achievements and perspectives. *ISPRS Journal of Photogrammetry and Remote Sensing*, 187: 306–327, 2022. 1

[27] Tsung-Yi Lin, Piotr Dollár, Ross Girshick, Kaiming He, Bharath Hariharan, and Serge Belongie. Feature Pyramid Networks for Object Detection. In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 936–944, 2017. 6

[28] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *2017 IEEE International Conference on Computer Vision (ICCV)*, pages 2999–3007, 2017. 6

[29] Denghong Liu, Xiaolin Zhu, Meredith Holgerson, Sheel Bansal, and Xiangtao Xu. Inventorying ponds through novel size-adaptive object mapping using Sentinel-1/2 time series. *Remote Sensing of Environment*, 315:114484, 2024. 3

[30] Jie Ma, Chuan Wang, Yang Liu, Liang Lin, and Guanbin Li. Enhanced soft label for semi-supervised semantic segmentation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 1185–1195, 2023. 3

[31] SM Nuruzzaman Manik, Georgina Pengilley, Geoffrey Dean, Brian Field, Sergey Shabala, and Meixue Zhou. Soil and crop management practices to minimize the impact of waterlogging on crop productivity. *Frontiers in plant science*, 10:140, 2019. 1

[32] T Mayer, A Poortinga, B Bhandari, A P Nicolau, K Markert, N S Thwal, A Markert, A Haag, J Kilbride, F Chishtie, A Wadhwa, N Clinton, and D Saah. Deep learning approach for Sentinel-1 surface water mapping leveraging Google Earth Engine. *ISPRS Open Journal of Photogrammetry and Remote Sensing*, 2:100005, 2021. 4

[33] Szabolcs Mihály, Gábor Remetey-Fülöpp, Dániel Kristóf, Anna Czinkóczky, Tamás Palya, László Pásztor, Pál Rudan, György Szabó, and László Zentai. Earth observation and geospatial big data management and engagement of stakeholders in Hungary to support the SDGs. *Big Earth Data*, 5 (3):306–351, 2021. 4

[34] Fabio Montello, Edoardo Arnaudo, and Claudio Rossi. Mmflood: A multimodal dataset for flood delineation from satellite imagery. *IEEE Access*, 10:96774–96787, 2022. 3

[35] Riju Mukherjee, Danish Ahmed, and Mujahir H. Abbasi. Detection of Surface Water from Satellite Imagery Using Deep Learning with Indirect Proxy Based Label Collection Method. In *2020 International Symposium on Advanced Electrical and Communication Technologies (ISAECT)*, pages 1–5, 2020. 3

[36] G. Nádor, G. Friedl, A. Rotterné Kulcsár, I. Hubik, V. Pacskó, and Gy. Surek. Agricultural damage mapping in hungary. In *The ever-growing use of COPERNICUS across Europe's regions*, pages 44–45. EC, ESA and NEREUS, 2018. 4

[37] Ozan Oktay, Jo Schlemper, Loic Folgoc, Matthew Lee, Mattias Heinrich, Kazunari Misawa, Kensaku Mori, Steven McDonagh, Nils Hammerla, Bernhard Kainz, Ben Glocker, and Daniel Rueckert. Attention U-Net: Learning Where to Look for the Pancreas. 2018. 6

[38] OPENGIS.ch. QField for QGIS. https://github.com/opengisch/QField, 2025. Version 3.6.5 (Gondwana), released June 14, 2025; accessed 2025-06-16. 4

[39] Wonpyo Park, Dongju Kim, Yan Lu, and Minsu Cho. Relational knowledge distillation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 3967–3976, 2019. 3

[40] Arash Modaresi Rad, Jason Kreitler, and Mojtaba Sadegh. Augmented Normalized Difference Water Index for improved surface water monitoring. *Environmental Modelling & Software*, 140:105030, 2021. 3

[41] Maryam Rahnemoonfar, Tashnim Chowdhury, Argho Sarkar, Debvrat Varshney, Masoud Yari, and Robin Roberson Murphy. Floodnet: A high resolution aerial imagery dataset for post flood scene understanding. *IEEE Access*, 9:89644–89654, 2021. 3

[42] C Rambour, N Audebert, E Koeniguer, B Le Saux, M Crucianu, and M Datcu. Sen12-flood: a sar and multispectral dataset for flood detection. *IEEE: Piscataway, NJ, USA*, 2020. 2

[43] He Ren, Zhen Yang, Fashuai Li, Maoxin Zhang, Yuwei Chen, and Tingting He. Satellite images reveal rapid development of global water-based photovoltaic over the past 20 years. *International Journal of Applied Earth Observation and Geoinformation*, 136:104354, 2025. 3

[44] Adriana Romero, Nicolas Ballas, Samira Ebrahimi Kahou, Antoine Chassang, Carlo Gatta, and Yoshua Bengio. Fitnets: Hints for thin deep nets. *arXiv preprint arXiv:1412.6550*, 2014. 3

[45] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional Networks for Biomedical Image Segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, pages 234–241, Cham, 2015. Springer International Publishing. 6

[46] J. W. Rouse, R. H. Haas, J. A. Schell, and D. W. Deering. Monitoring vegetation systems in the Great Plains with ERTS. In *Proceedings of the Third Earth Resources Technology Satellite-1 Symposium, Volume I: Technical Presentations*, pages 309–317. NASA SP-351, 1974. 4

[47] Subham Roy, Arghadeep Bose, Nimai Singha, Debanjan Basak, and Indrajit Roy Chowdhury. Urban waterlogging risk as an undervalued environmental challenge: An Integrated MCDA-GIS based modeling approach. *Environmental Challenges*, 4:100194, 2021. 1

[48] Maha Shadaydeh, András Zlinszky, Andrea Manno-Kovacs, and Tamas Sziranyi. Wetland mapping by fusion of airborne laser scanning and multi-temporal multispectral satellite imagery. *International Journal of Remote Sensing*, 38(23): 7422–7440, 2017. 3

[49] J Sheffield, Eric F Wood, M Pan, H Beck, G Coccia, A Serrat-Capdevila, and KJWRR Verbist. Satellite remote sensing for water resources management: Potential for sup-

porting sustainable development in data-poor regions. *Water Resources Research*, 54(12):9724–9758, 2018. 1

[50] Suqi Song, Chenxu Zhang, Peng Zhang, Pengkun Li, Fenglong Song, and Lei Zhang. Urban Waterlogging Detection: A Challenging Benchmark and Large-Small Model Co-adapter. In *ECCV*, pages 366–382. Springer, 2024. 3

[51] Carole H. Sudre, Wenqi Li, Tom Vercauteren, Sebastien Ourselin, and M. Jorge Cardoso. Generalised dice overlap as a deep learning loss function for highly unbalanced segmentations. In *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support*, pages 240–248, Cham, 2017. Springer International Publishing. 6

[52] Tamas Sziranyi and Maha Shadaydeh. Segmentation of Remote Sensing Images Using Similarity-Measure-Based Fusion-MRF Model. *IEEE Geoscience and Remote Sensing Letters*, 11(9):1544–1548, 2014. 3

[53] Li-xin Tian, Yu-chuan Zhang, Peng-liang Chen, Fei-fei Zhang, Jing Li, Feng Yan, Yang Dong, and Bai-li Feng. How does the waterlogging regime affect crop yield? A global meta-analysis. *Frontiers in plant science*, 12:634898, 2021. 1

[54] Nikhil Kumar Tomar, Ulas Bagci, and Debesh Jha. RUPNet: Residual upsampling network for real-time polyp segmentation. In *Medical Imaging 2023*. SPIE, 2023. Publisher Copyright: © 2023 SPIE.; Medical Imaging 2023: Computer-Aided Diagnosis. 6

[55] Viktoriya Tsyganskaya, Sandro Martinis, Philip Marzahn, and Ralf Ludwig. Detection of temporary flooded vegetation using Sentinel-1 time series data. *Remote Sensing*, 10 (8):1286, 2018. 3

[56] European Union. REGULATION (EU) 2021/2115 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL, 2021. Accessed: 2025-05-22. 1

[57] European Union. REGULATION (EU) 2024/1991 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL, 2024. Accessed: 2025-05-22. 1

[58] Guoqing Wang, Ning Zhang, Jue Wang, Wenchao Liu, Yizhuang Xie, and He Chen. Knowledge Distillation-Based Lightweight Change Detection in High-Resolution Remote Sensing Imagery for On-Board Processing. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 17:3860–3877, 2024. 3

[59] Shuai Wang, Yu Chen, Yafei Yuan, Xinlong Chen, Jinze Tian, Xiaolong Tian, and Huibin Cheng. Tsae-unet: A novel network for multi-scene and multi-temporal water body detection based on spatiotemporal feature extraction. *Remote Sensing*, 16(20):3829, 2024. 3

[60] Wieland, Marc and Fichtner, Florian and Martinis, Sandro and Groth, Sandro and Krullikowski, Christian and Plank, Simon and Motagh, Mahdi. S1s2-water: A global dataset for semantic segmentation of water bodies from sentinel- 1 and sentinel-2 satellite images. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 17:1084–1099, 2024. 3

[61] Michael Wurm, Thomas Stark, Xiao Xiang Zhu, Matthias Weigand, and Hannes Taubenböck. Semantic segmentation of slums in satellite images using transfer learning on fully convolutional neural networks. *ISPRS Journal of Photogrammetry and Remote Sensing*, 150:59–69, 2019. 3

[62] Liping Yang, Joshua Driscol, Sarigai Sarigai, Qiusheng Wu, Christopher D Lippitt, and Melinda Morgan. Towards synoptic water monitoring systems: a review of AI methods for automating water body detection and water quality monitoring using remote sensing. *Sensors*, 22(6):2416, 2022. 1

[63] Qing Yang, Xinyi Shen, Emmanouil N Anagnostou, Chongxun Mo, Jack R Eggleston, and Albert J Kettner. A high-resolution flood inundation archive (2016–the present) from Sentinel-1 SAR imagery over CONUS. *Bulletin of the American Meteorological Society*, 102(5):E1064–E1079, 2021. 3

[64] Zi-An Yang, Nai-Rong Zheng, Xian-Zheng Shi, Wen-Bo Zhou, and Feng Wang. Precise and Fast Segmentation of Sea Ice in High-Resolution Images Based on Multiscale and Knowledge Distillation. In *IGARSS 2023 - 2023 IEEE International Geoscience and Remote Sensing Symposium*, pages 4946–4949, 2023. 3

[65] Jibo Yue, Jia Tian, Qingjiu Tian, Kaijian Xu, and Nianxu Xu. Development of soil moisture indices from differences in water absorption between shortwave-infrared bands. *ISPRS Journal of Photogrammetry and Remote Sensing*, 154: 216–230, 2019. 4

[66] Jingting Zhang, Jiantao Gao, Jinshuo Liang, Yiqiang Wu, Bin Li, Yang Zhai, and Xiaomao Li. Efficient Water Segmentation with Transformer and Knowledge Distillation for USVs. *Journal of Marine Science and Engineering*, 11(5), 2023. 3

[67] Linfeng Zhang, Jiebo Song, Anni Gao, Jingwei Chen, Chenglong Bao, and Kaisheng Ma. Be Your Own Teacher: Improve the Performance of Convolutional Neural Networks via Self Distillation. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 3712–3721, 2019. 6

[68] Pei Zhang, Ying Li, Dong Wang, and Jiyue Wang. RS-SSKD: Self-Supervision Equipped with Knowledge Distillation for Few-Shot Remote Sensing Scene Classification. *Sensors*, 21(5), 2021. 3

[69] Hengshuang Zhao, Jianping Shi, Xiaojuan Qi, Xiaogang Wang, and Jiaya Jia. Pyramid Scene Parsing Network. In *CVPR*, 2017. 6

[70] Jie Zhao, Zhitong Xiong, and Xiao Xiang Zhu. UrbanSARFloods: Sentinel-1 SLC-Based Benchmark Dataset for Urban and Open-Area Flood Mapping. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 419–429, 2024. 3

[71] Wujie Zhou, Xiaomin Fan, Weiqing Yan, Shengdao Shan, Qiuping Jiang, and Jenq-Neng Hwang. Graph Attention Guidance Network With Knowledge Distillation for Semantic Segmentation of Remote Sensing Images. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–15, 2023. 3

[72] Zongwei Zhou, Md Mahfuzur Rahman Siddiquee, Nima Tajbakhsh, and Jianming Liang. UNet++: A Nested U-Net Architecture for Medical Image Segmentation. In *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support*, pages 3–11, 2018. 6